

Universitatea
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București



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Automatică și
Calculatoare



Catedra de
Calculatoare

Database Connection

Autor

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Cuprins

- Java + Oracle
- Python + Oracle
- C# + Microsoft SQL Server



Pachet Oracle

```
create or replace package pck_example
as
  procedure exemplu_procedura(idDept in dept.deptno%type,
        numeDept out dept.dname%type, angajati out sys_refcursor);
  function exemplu_functie(gradSal in salgrade.grade%type,
        salLow out salgrade.losal%type, salHigh out salgrade.hisal%type) return number;
end pck_example;
/
```



Pachet Oracle

```
create or replace package body pck_example
as
  function exemplu_functie(gradSal in salgrade.grade%type,
        salLow out salgrade.losal%type, salHigh out salgrade.hisal%type) return number
  is
    avgVenit number;
  begin
    select distinct losal, hisal into salLow, salHigh from salgrade where grade = gradSal;
    select round(avg(sal+nvl(comm, 0)), 2) into avgVenit
        from emp where sal + nvl(comm, 0) between salLow and salHigh;
    return avgVenit;
  end exemplu_functie;

  procedure exemplu_procedura(idDept in dept.deptno%type,
        numeDept out dept.dname%type, angajati out sys_refcursor)
  is
  begin
    select dname into numeDept from dept where deptno = iddept;
    open angajati for
        select ename, hiredate, job, sal+nvl(comm, 0) venit
        from emp where deptno = idDept;
  end exemplu_procedura;
end pck_example;
/
```



Java + Oracle

- Pas I – se descarcă Java JDK
- Pas II – se descarcă Oracle JDBC Driver
- Pas III – în IDE-ul preferat de dezvoltator, se creează un nou proiect
- Pas IV – se adaugă referință către Oracle JDBC Driver



Java + Oracle

- Java JDK
  - http://www.oracle.com/technetwork/java/javase/downloads/index.html
- Oracle JDBC driver
  - http://www.oracle.com/technetwork/database/features/jdbc/index-091264.html
- Tutorials
  - https://docs.oracle.com/database/121/JJDEV/toc.htm
  - https://docs.oracle.com/database/121/JJDBC/toc.htm
  - https://docs.oracle.com/database/121/JAJDB/toc.htm



Java + Oracle

- Se creează o clasă pentru conexiunea cu baza de date

```
package oracleexample;

import java.sql.CallableStatement;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.ResultSet;

public class OracleConnection {

    private Connection connection = null;
    private String username = null;
    private String password = null;
    private String host = null;
    private int port;
    private String schema = null;
```



Java + Oracle

- Constructor

```
public OracleConnection(String host, int port, String schema,
        String username, String password) {
    this.host = host;
    this.port = port;
    this.schema = schema;
    this.username = username;
    this.password = password;
    try {
        Class.forName("oracle.jdbc.driver.OracleDriver");
        System.out.println("JDBC check: PASSED!");
    } catch (ClassNotFoundException CNFEx) {
        System.out.println("JDBC check: NOT PASSED!");
        CNFEx.printStackTrace();
    }
}
```



Java + Oracle

- Conectarea la baza de date

```
public void openConnection() {
    try {
        String connString = "jdbc:oracle:thin:@" + this.host + ":" +
                this.port + ":" + this.schema;
        this.connection = DriverManager.getConnection(connString,
                this.username, this.password);
        if (this.connection != null) {
            System.out.println("We are connected!");
        } else {
            System.out.println("Failed to make a connection!");
        }
    } catch (SQLException e) {
        System.out.println("Connection Failed! Check output console");
        e.printStackTrace();
    }
}
```



Java + Oracle

- Deconectarea de la baza de date

```
public void closeConnection() {
    try {
        this.connection.close();
        System.out.println("Connection closed!");
    } catch (Exception e) {
        System.out.println("Connection NOT closed!");
        e.printStackTrace();
    }
}
```



Java + Oracle

- Folosirea unei proceduri

```
public void exemplu_procedura(int idDept) {
    try {
        String call = "{call pck_example.exemplu_procedura(?,?, ?)}";
        CallableStatement cstmt = this.connection.prepareCall(call);
        cstmt.setInt("idDept", idDept);
        // register the out paremeters
        cstmt.registerOutParameter("numeDept", oracle.jdbc.OracleTypes.VARCHAR);
        cstmt.registerOutParameter("angajati", oracle.jdbc.OracleTypes.CURSOR);
        cstmt.executeQuery();
        // get the department's name
        String numeDept = cstmt.getString("numeDept");
        // get the cursor
        ResultSet rs = (ResultSet) cstmt.getObject("angajati");
        while (rs.next()) {
            System.out.println(rs.getString("ename") + "\t" + rs.getString("hiredate") + "\t" +
                    rs.getString("job") + "\t" + rs.getInt("venit") + "\t" + numeDept);
        }
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
}
```



Java + Oracle

- Folosirea unei funcții

```
public void exemplu_functie(int gradSal) {
    try {
        String call = "{? = call pck_example.exemplu_functie(?, ?, ?)}";
        CallableStatement cstmt = this.connection.prepareCall(call);
        // register parameters
        cstmt.registerOutParameter(1, oracle.jdbc.OracleTypes.NUMBER);
        cstmt.setInt(2, gradSal);
        cstmt.registerOutParameter(3, oracle.jdbc.OracleTypes.NUMBER);
        cstmt.registerOutParameter(4, oracle.jdbc.OracleTypes.NUMBER);
        cstmt.executeQuery();
        // get the department's name
        Float avgVenit = cstmt.getFloat(1);
        Float salLow = cstmt.getFloat(3);
        Float salHigh = cstmt.getFloat(4);

        System.out.println(avgVenit + "\t" + salLow + "\t" + salHigh);
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
```



Java + Oracle

- Utilizare

```
public class OracleExample {
    public static void main(String[] args) {
        OracleConnection oc = new OracleConnection("localhost", 1521,
                "ORCL", "student", "student");
        oc.openConnection();
        oc.exemplu_procedura(10);
        oc.exemplu_functie(3);
        oc.closeConnection();
    }
}
```



Python + Oracle

- Pas I – se instalează Python >= v2.7.8
- Pas II – se instalează pachetul cx_Oracle

- Configure & Installing cx_Oracle Windows
  - pip install cx_Oracle
- Configure & Installing cx_Oracle Linux (Redhat distributions)
  - https://blogs.oracle.com/opal/entry/configuring_python_cx_oracle_and
- Documentation
  - http://cx-oracle.readthedocs.org/en/latest/index.html
- Tutorials
  - http://www.oracle.com/webfolder/technetwork/tutorials/obe/db/OOW11/python_db/python_db.htm
  - http://www.oracle.com/technetwork/articles/dsl/mastering-oracle-python-1391323.html



Python + Oracle

- Se creează o clasă pentru conexiunea cu baza de date

```
import cx_Oracle

class OracleConnection:

    def __init__(self, host, port, schema, username, password):
        self.host = host
        self.port = port
        self.schema = schema
        self.username = username
        self.password = password
```



Python + Oracle

- Conectarea la baza de date

```
def openConnection(self):
    try:
        connString = self.username + "/" + self.password
        connString += "@//" + self.host + ":" + str(self.port)
        connString += "/" + self.schema
        self.db = cx_Oracle.Connection(connString)
        self.cursor = self.db.cursor()
        print "Connection open!"
    except Exception, e:
        print "Connection not open!"
        print e
```



Python + Oracle

- Deconectarea de la baza de date

```
def closeConnection(self):
    try:
        self.cursor.close()
        self.db.close()
        print "Connection close!"
    except Exception, e:
        print "Connection not closed!"
        print e
```



Python + Oracle

- Folosirea unei proceduri

```
def exempluProcedura(self, idDept):
    try:
        numeDept = self.cursor.var(cx_Oracle.STRING)
        angajati = self.cursor.var(cx_Oracle.CURSOR)
        self.cursor.callproc("pck_example.exemplu_procedura", [idDept, numeDept, angajati])
        print numeDept.getvalue()
        for elem in angajati.getvalue():
            print elem[0], elem[1].strftime("%Y-%m-%d"), elem[2], elem[3], numeDept.getvalue()
    except Exception, e:
        print e
```



Python + Oracle

- Folosirea unei funcții

```
def exempluFunctie(self, salGrad):
    try:

        salLow = self.cursor.var(cx_Oracle.NUMBER)
        salHigh = self.cursor.var(cx_Oracle.NUMBER)
        avgVenit = self.cursor.callfunc("pck_example.exemplu_functie", cx_Oracle.NUMBER, [salGrad, salLow, salHigh])
        print avgVenit, salLow.getvalue(), salHigh.getvalue()
    except Exception, e:
        print e
```



Python + Oracle

- Utilizare

```
if __name__ == "__main__":
    oc = OracleConnection('localhost', 1521, 'orcl', 'student', 'student')
    oc.openConnection()
    oc.exempluProcedura(10)
    oc.exempluFunctie(2)
    oc.closeConnection()
```



Procedură Microsoft SQL Server

```
create procedure exemplu_procedura(
  @idDept int,
  @numeDept varchar(10) output)
as
begin
  select @numeDept = dname from dept where deptno = @idDept;
  select ename, hiredate, job, sal+isnull(comm, 0) venit
      from emp where deptno = @idDept;
end;
```



Funcție Microsoft SQL Server

```
create function exemplu_functie(@gradSal int) returns real
as
begin
    declare @avgVenit real
    declare @salLow real
    declare @salHigh real
    select @salLow = losal, @salHigh = hisal
        from salgrade where grade = @gradSal
    select @avgVenit = round(avg(sal+isnull(comm, 0)), 2)
        from emp where sal + isnull(comm, 0) between @salLow and @salHigh
    return @avgVenit
end
```



C# + Microsoft SQL Server

- Pas I – Instalare Microsoft SQL Server
- Pas II – Instalare Microsoft Visual Studio

- Tutorial C#
  - https://msdn.microsoft.com/ro-ro/library/aa288436(v=vs.71).aspx
- Tutorial MS SQL Server
  - https://msdn.microsoft.com/en-us/library/ms167593(v=sql.105).aspx



C# + Microsoft SQL Server

- Librării folosite

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Data.SqlClient;
using System.Data;
```



C# + Microsoft SQL Server

```
namespace MSSQLExample
{
    class MSSQLConnection
    {
        private string host;
        private int port;
        private string dbname;
        private string username;
        private string password;
        private SqlConnection conn;
        private SqlCommand com;

        public MSSQLConnection(string host, int port, string dbname,
                string username, string password)
        {
            this.host = host;
            this.port = port;
            this.dbname = dbname;
            this.username = username;
            this.password = password;
            string connString = "Data Source=" + this.host +
                ", " + this.port + ";Initial Catalog=" + this.dbname +
                ";User ID=" + this.username + ";Password=" + this.password;
            this.conn = new SqlConnection(connString);
        }
```



C# + Microsoft SQL Server

```
public void exempluProcedura(int idDept)
{
    try
    {
        this.conn.Open();
        this.com = new SqlCommand("exemplu_procedura", this.conn);
        this.com.CommandType = CommandType.StoredProcedure;

        SqlParameter idDeptParam = new SqlParameter("@idDept", SqlDbType.Int);
        idDeptParam.Direction = ParameterDirection.Input;
        idDeptParam.Value = idDept;

        SqlParameter numeDeptParam = new SqlParameter("@numeDept", SqlDbType.VarChar, 10);
        numeDeptParam.Direction = ParameterDirection.Output;

        this.com.Parameters.Add(idDeptParam);
        this.com.Parameters.Add(numeDeptParam);
        SqlDataAdapter adapter = new SqlDataAdapter(this.com);

        DataTable dt = new DataTable();
        adapter.Fill(dt);
```



C# + Microsoft SQL Server

```
            string numeDept = Convert.ToString(com.Parameters["@numeDept"].Value);
            if (dt.Rows.Count > 0)
            {
                foreach (DataRow dr in dt.Rows)
                {
                    Console.WriteLine(dr["ename"] + "\t" + dr["hiredate"]
                        + "\t" + dr["job"] + "\t" + dr["venit"] + "\t" + numeDept);
                }
            }
        }
        catch (Exception exp)
        {
            Console.WriteLine("Eroare Procedura!\r\n" + exp.ToString());
        }
        finally
        {
            this.com.Parameters.Clear();
            this.com.Dispose();
            this.conn.Close();
        }
    }
```



C# + Microsoft SQL Server

```
public void exempluFunctie(int grad)
{
    try
    {
        this.conn.Open();
        this.com = new SqlCommand("exemplu_functie", this.conn);
        this.com.CommandType = CommandType.StoredProcedure;

        SqlParameter gradSalParam = new SqlParameter("@gradSal", SqlDbType.Int);
        gradSalParam.Direction = ParameterDirection.Input;
        gradSalParam.Value = grad;

        SqlParameter avgVenitParam = new SqlParameter("@Result", SqlDbType.Real);
        avgVenitParam.Direction = ParameterDirection.ReturnValue;

        this.com.Parameters.Add(gradSalParam);
        this.com.Parameters.Add(avgVenitParam);

        this.com.ExecuteNonQuery();

        if (avgVenitParam.Value != DBNull.Value)
        {
            Console.WriteLine("Venit mediu:" + avgVenitParam.Value);
        }
    }
```



C# + Microsoft SQL Server

```
            catch (Exception exp)
            {
                Console.WriteLine("Eroare Functie!\r\n" + exp.ToString());
            }
            finally
            {
                this.com.Parameters.Clear();
                this.com.Dispose();
                this.conn.Close();
            }
        }
    }
}
```



C# + Microsoft SQL Server

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace MSSQLExample
{
    class Program
    {
        static void Main(string[] args)
        {
            MSSQLConnection mssqlconn = new MSSQLConnection("localhost", 1433, "hr", "sa", "123456");
            mssqlconn.exempluProcedura(10);
            mssqlconn.exempluFunctie(3);
            Console.Read();
        }
    }
}
```